Bayesian Credit Risk Particle Filter

CASE STUDY

ENVIRONMENT SETUP

Synthetic portfolio of 500 Debtors
Portfolio Value: $ 11.75 Million

Model employs three debtor-specific features for the Logistic Regression model.

Debtors range across five exposure classes:

- Retail [0]
- Corporate [1]
- Institutions [2]
- Sovereign [3]
- Real Estate [4]

id	feature_1	feature_2	feature_3	EAD	exposure_class
1	3	4	2	12450	0
2	2	3	5	18920	1
3	5	6	1	25330	2
4	4	5	3	31200	3
5	1	2	4	9870	0
6	3	4	2	14560	1
7	4	5	3	22140	2
8	2	3	1	28450	3
9	5	6	4	35670	4
10	3	4	2	11230	0
11	4	5	3	19480	1
12	2	3	1	26750	2
13	5	6	4	33920	3
14	1	2	5	8940	0
15	3	4	2	15680	1
16	4	5	3	23450	2
17	5	6	1	30120	4
18	2	3	4	37890	0
19	3	4	2	13670	1
20	4	5	3	21340	2
21	5	6	1	28560	3
22	1	2	4	36780	4
23	3	4	5	10980	0
24	2	3	2	18230	1
25	4	5	3	25640	2
26	5	6	1	32870	3

PARTICLE GRID

GDP	Unemployment	Inflation	Interest rates	Oil prices
-4.72235	6.11821	-0.317867	1.09608	61.2229
2.67142	2.26244	4.40392	5.81787	125.103
7.55775	4.30557	0.629204	4.8708	75.0759
5.45545	3.31819	1.75992	1.43237	63.9717
3.50759	4.72911	2.30883	6.97836	80.9577
1.27604	2.67106	0.0134304	4.295	115.578
4.90392	5.62244	0.776044	6.76047	19.6742
-1.46052	6.42649	4.23319	3.05328	85.6763
-0.978356	4.19461	-0.14981	1.59344	119.834
3.94511	7.61104	4.58056	6.35868	59.8408
5.26404	5.03219	3.29709	8.37038	48.278
4.56763	4.62575	1.62229	4.43635	54.2076
2.25992	2.47685	3.90656	7.21063	25.3948
3.3655	3.72039	0.479911	3.32858	65.3427
-0.455453	7.03558	1.69117	1.18213	50.5489
2.05328	5.37756	-0.764043	3.53427	72.95
5.76047	6.16882	2.795	7.59338	128.346
3.72396	6.58578	-0.0676273	4.08531	91.7139
2.60283	7.54191	1.1345	2.98408	29.0748
4.02009	3.90486	4.31787	4.94511	70.1592
-1.87847	3.38896	3.6722	4.3655	158.102
6.21063	5.18091	1.41469	7.09252	88.1787
-0.080564	4.36017	2.51592	3.25992	24.0662
-0.264043	8.73756	3.52009	8.55775	63.2858
1.05489	4.02312	4.85868	6.45545	52.0291
1.6345	6.91898	0.702909	3.39717	80.2009

Particle grid generated through Quasi-Monte Carlo Sequences. Each particle is a 5-dimensional vector with each dimension as a macro-economic state variable.

Total generated particles: 985

Macro-economic state variables used:

- GDP
- Unemployment
- Inflation
- Interest rates
- Oil prices

```
struct Particle
{
    static constexpr int n_features = 5;

    union
    {
        struct
        {
            double gdp;       // GDP growth rate (%)
            double unemp;     // Unemployment rate (%)
            double inflation; // Inflation rate (%)
            double interest;  // Interest rate (%)
            double oilPrice;  // Oil price change (%)
        };
        double features[n_features];
    };
};
struct Debtor
{

    static constexpr int n_features = 3;
    static constexpr int n_exposure_classes = 5;

    int id;
    union
    {
        struct
        {
            double feature_1;
            double feature_2;
            double feature_3;
        };
        double features[n_features];
    };
    double EAD;         // Exposure at Default
    int exposure_class; // 0: retail, 1: corporate, 2: institutions, 3: realEstate, 4: sovereign
};
```

```
// {GDP, Unemployment, Inflation, InterestRates, OilPrices}
inline const double SystematicFactorSensitivities[][Particle::n_features] = {
    {-0.40, 0.35, 0.10, 0.03, -0.005}, // RETAIL
    {-0.35, 0.40, 0.15, 0.04, -0.010}, // CORPORATE
    {-0.20, 0.20, 0.08, 0.03, -0.005}, // INSTITUTIONS
    {-0.50, 0.45, 0.28, 0.08, -0.005}, // REALESTATE
    {-0.55, 0.50, 0.12, 0.02, 0.020},  // SOVEREIGN
};
// {feature_1, feature_2, feature_3}
inline const double IdiosyncraticFactorSensitivities[][Debtor::n_features] = {
    {0.08, -0.05, 0.12}, // RETAIL
    {0.11, -0.03, 0.09}, // CORPORATE
    {0.06, -0.02, 0.05}, // INSTITUTIONS
    {0.14, -0.06, 0.11}, // REALESTATE
    {0.05, -0.01, 0.04}, // SOVEREIGN
};
```

IMPLEMENTATION LANGUAGES USED:

C++ for Logic

Python for Risk Analysis

INITIALIZATION

The model is initialized with equal weights for each particle, which self-correct on observing default events. Particles are also segregated based on regimes. A regime's weight is the summation of weights of all the particles belonging to that regime.

We'll consider two cases for analysis:

- 2 defaults observed
- 7 defaults observed

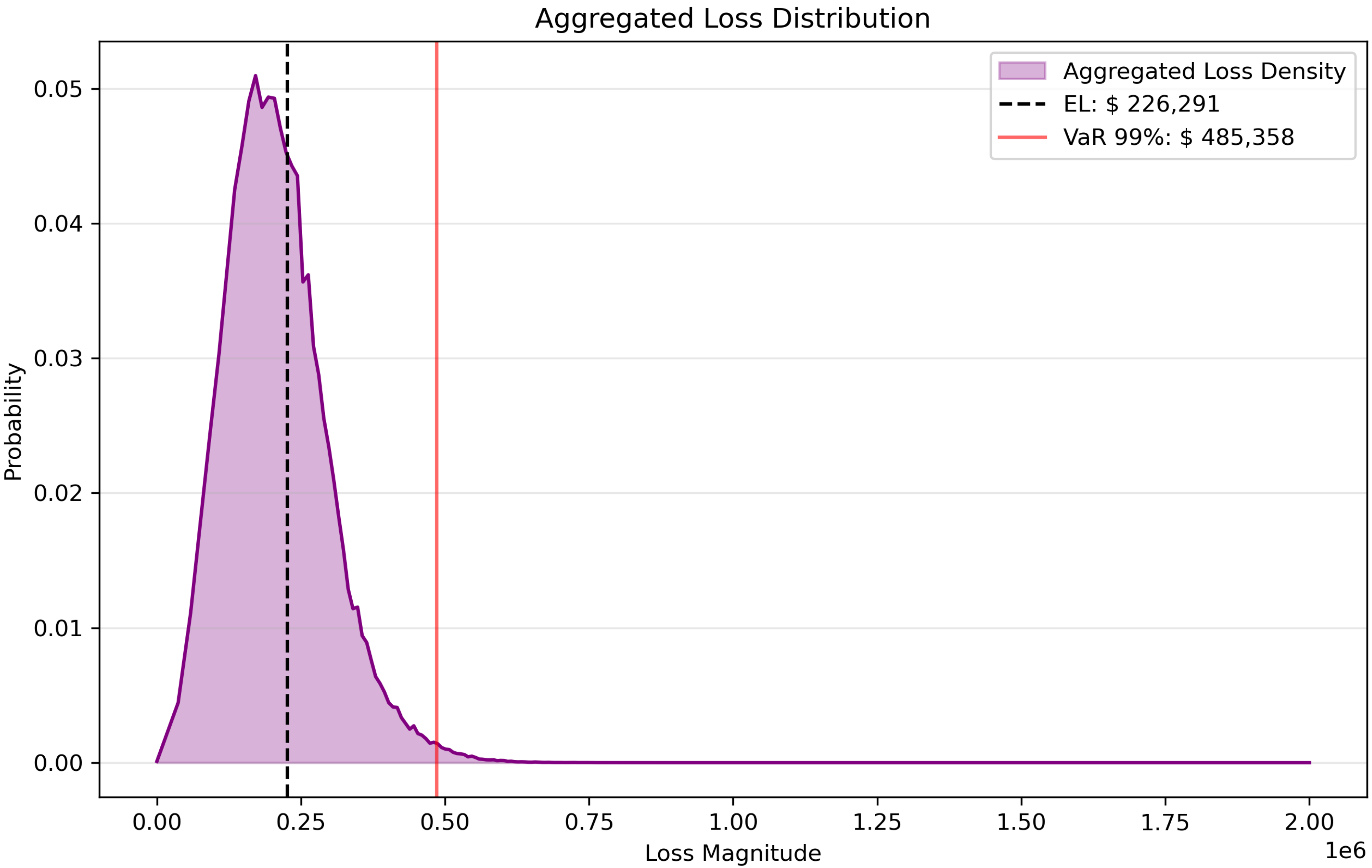
Loaded 985 Weights, 985 PD sets, 2 this month's defaults.

```
Weight update complete!
Total particles: 985

Regime Weight Distribution:
-------------------------------------------------------
Crisis     | # (0.012)
Recession  | ################ (0.184)
Normal     | ################################ (0.362)
Expansion  | ######################################## (0.442)
-------------------------------------------------------

Aggregated distribution:
Expected Loss: $ 226,291
Unexpected Loss: $ 90,188
Expected Shortfall (99%): $ 532,944
VaR (99%): $ 485,358
VaR (95%): $ 386,417
Economic Capital: $ 370,028
Tail Ratio (VaR_99.9 / VaR_99): 1.229

Regime wise Distributions:
```

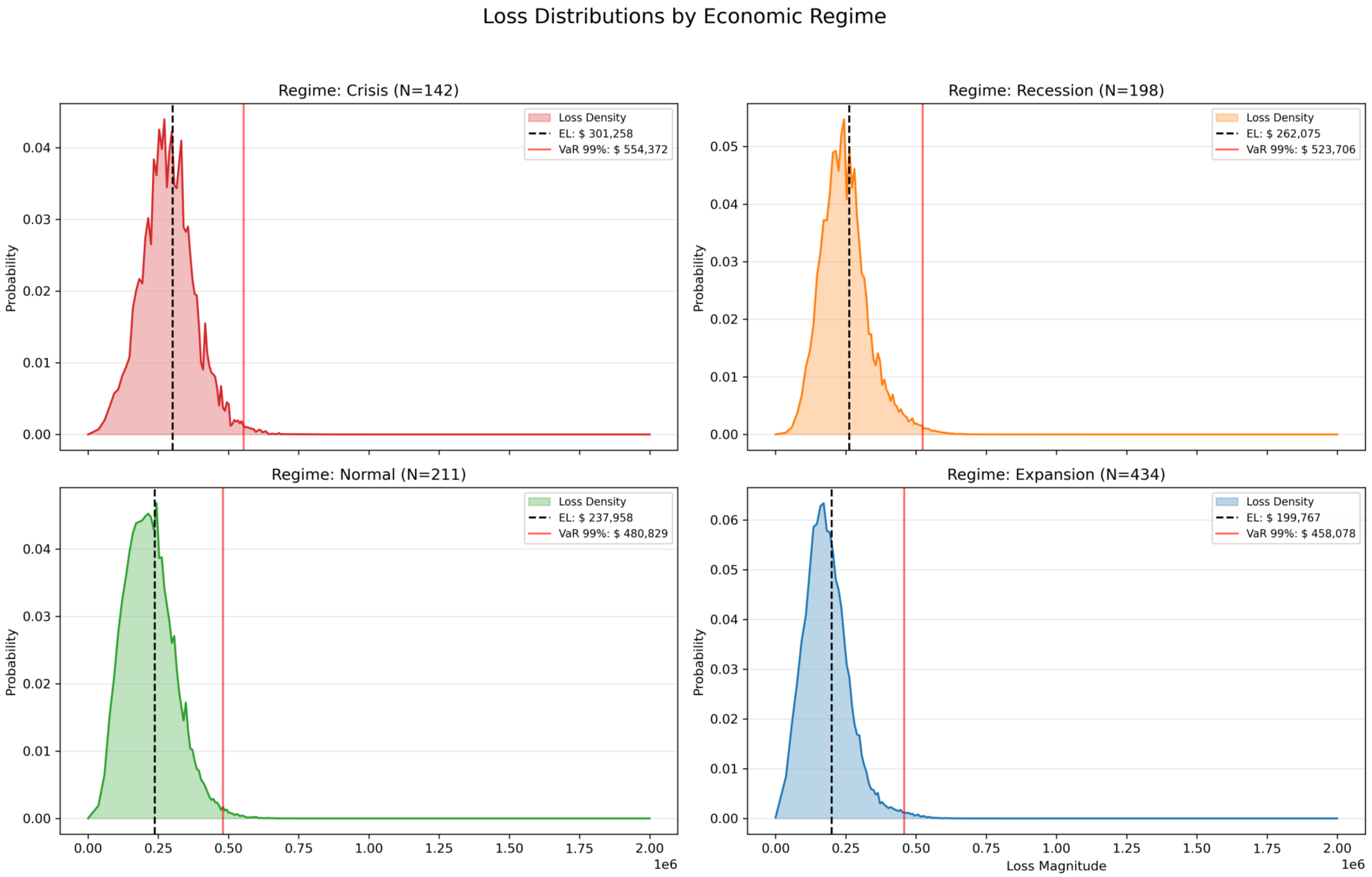
	Crisis	Recession	Normal	Expansion
Expected Loss	$ 301,258	$ 262,075	$ 237,958	$ 199,767
Unexpected Loss	$ 94,271	$ 88,281	$ 90,134	$ 82,210
Expected Shortfall (99%)	$ 608,938	$ 573,633	$ 533,336	$ 503,886
VaR (99%)	$ 554,372	$ 523,706	$ 480,829	$ 458,078
VaR (95%)	$ 459,776	$ 423,909	$ 393,307	$ 343,479
Economic Capital	$ 374,882	$ 369,275	$ 356,336	$ 359,544
Tail Ratio (VaR_99.9 / VaR_99)	1.220	1.206	1.236	1.221

CASE 1

2 Defaults observed



Aggregated Loss Distribution
Aggregated Loss Density
EL: $ 226,291
VaR 99%: $ 485,358
Probability
0.05
0.04
0.03
0.02
0.01
0.00
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
1e6
Loss Magnitude



Loss Distributions by Economic Regime
Regime: Crisis (N=142)
Loss Density
EL: $ 301,258
VaR 99%: $ 554,372
Regime: Recession (N=198)
Loss Density
EL: $ 262,075
VaR 99%: $ 523,706
Regime: Normal (N=211)
Loss Density
EL: $ 237,958
VaR 99%: $ 480,829
Regime: Expansion (N=434)
Loss Density
EL: $ 199,767
VaR 99%: $ 458,078
Probability
Loss Magnitude
1e6

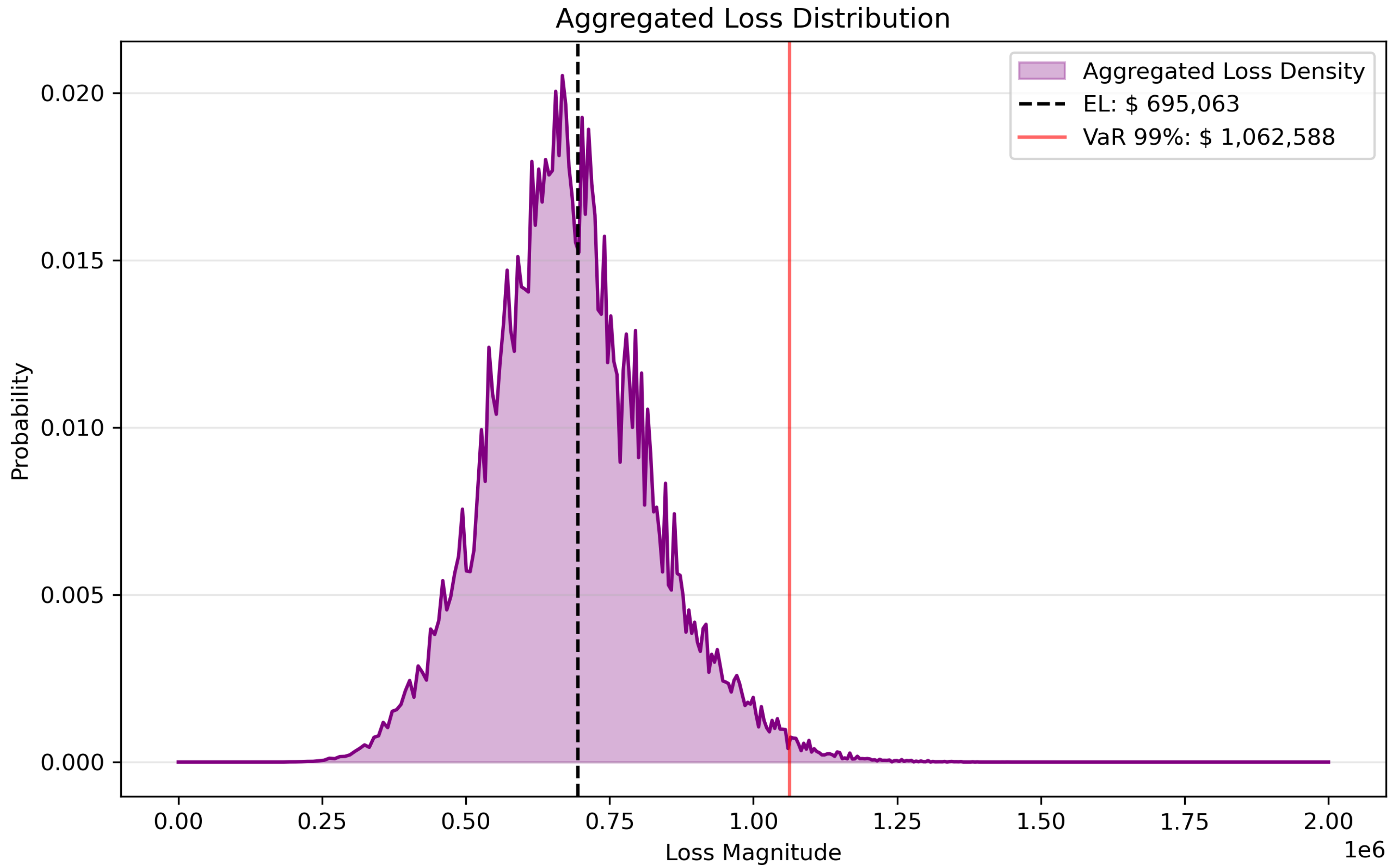
CASE 2

7 Defaults observed

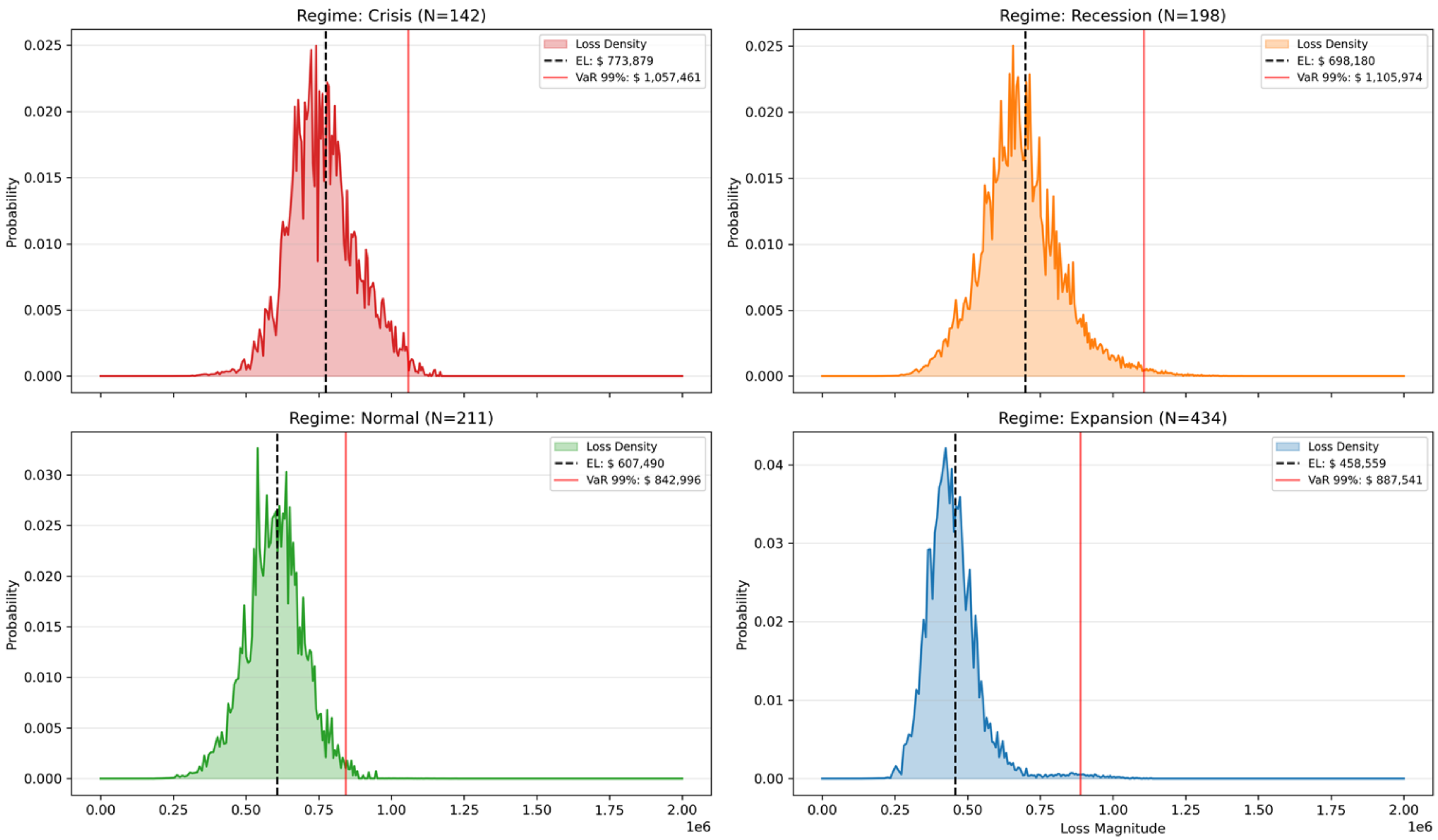
Loaded 985 Weights, 985 PD sets, 7 this month's defaults.

```
Weight update complete!
Total particles: 985

Regime Weight Distribution:
-------------------------------------------------------
Crisis     | ########################### (0.302)
Recession  | ######################################## (0.432)
Normal     | ####################### (0.255)
Expansion  | # (0.012)
-------------------------------------------------------

Aggregated distribution:
Expected Loss: $ 695,063
Unexpected Loss: $ 141,054
Expected Shortfall (99%): $ 1,126,634
VaR (99%): $ 1,062,588
VaR (95%): $ 942,866
Economic Capital: $ 515,784
Tail Ratio (VaR_99.9 / VaR_99): 1.140

Regime wise Distributions:
```

	Crisis	Recession	Normal	Expansion
Expected Loss	$ 773,879	$ 698,180	$ 607,490	$ 458,559
Unexpected Loss	$ 118,236	$ 140,900	$ 98,784	$ 101,380
Expected Shortfall (99%)	$ 1,093,890	$ 1,178,654	$ 882,330	$ 957,563
VaR (99%)	$ 1,057,461	$ 1,105,974	$ 842,996	$ 887,541
VaR (95%)	$ 983,984	$ 949,380	$ 772,344	$ 617,299
Economic Capital	$ 370,480	$ 566,737	$ 336,224	$ 572,631
Tail Ratio (VaR_99.9 / VaR_99)	1.082	1.144	1.119	1.162


Aggregated Loss Distribution
Aggregated Loss Density
EL: $ 695,063
VaR 99%: $ 1,062,588
Probability
0.020
0.015
0.010
0.005
0.000
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Loss Magnitude
1e6

Loss Distributions by Economic Regime



CONCLUSION

As illustrated, the model flagged increasing recession probabilities and elevated risk measures when seven defaults occurred. While we considered single-period events, the model may also have learned this trend over time.

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